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SUPERSEDING
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FEDERAL SPECIFICATION

CLOTH, COATED (RUBBER AND PLASTIC) AND PLASTIC SHEETING (ROLLS), FOR HOSPITAL USE

(Formerly "Sheeting; Rubber")

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—Hospital sheeting covered by this specification is intended for use in hospitals, for the protection of mattresses.

1.2 Classification.

1.2.1 Grade and types.—Hospital sheeting covered by this specification shall be of one grade and of the following types, as specified:

Type I.—Rubber, double coated fabric.

Type II.—Plastic, double coated fabric.

Type III.—Plastic, unsupported film.

1.2.2 *Width.*

1.2.2.1 *Types I and II.*—Types I and II sheeting shall be 36, 45 or 54 inches wide, as specified.

1.2.2.2 *Type III.*—Type III sheeting shall be 36½, 48, or 54 inches wide, as specified.

1.2.3 *Classes.*—Type III sheeting shall be of the following classes, as specified:

Class 1.—0.004 inch in thickness.

Class 2.—0.006 inch in thickness.

1.2.4 *Color.*

1.2.4.1 Type I sheeting shall be furnished in the following colors, as specified:

Black.
White.
Maroon.

1.2.4.2 Types II and III sheeting shall be furnished in the following colors, as specified:

Natural color—unpigmented.
Black.
White.
Maroon.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

CCC–T–191—Textile Test Methods.
PPP–B–585—Boxes; Wood, Wirebound.
PPP–B–591—Boxes, Fiberboard, Wood-Cleated.

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FSC 8305, 9330

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–636—Boxes, Fiber.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Test Method Std. No. 601—Rubber; Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier Materials, Waterproof, Flexible.

MIL–A–140 — Adhesive, Water-Resistant, Waterproof Barrier Material.

MIL–B–138—Boxes, Wood, Fiberboard-Lined for Overseas Shipment (for Weight of Contents not Exceeding 500 Pounds).

MIL–P–12439—Packaging and Packing of Nylon, Rayon, and Rayon-Cotton Fabrics.

Military Standard:

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications. — The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Freight Classification Rules.

(Copies of Consolidated Freight Classification Rules may be obtained from the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Fabric.*—The fabric for types I and II hospital sheeting shall be woven from cotton or any suitable synthetic fiber. The fabric shall have closely woven selvage edges not less than 1/8 inch in width. The fabric shall be uniformly woven and shall be free from lumps, knots, and other defects.

3.1.2 *Rubber.*—Type I sheeting shall be coated or impregnated with a rubber compound of natural rubber, synthetic rubber, or a mixture of the two. The rubber compound shall not contain any ingredient that may have an irritating effect on the skin. The compound shall not have a disagreeable odor.

3.1.3 *Plastic.*—The plastic compound used for coating or impregnating type II sheeting and in type III unsupported sheeting shall consist of a vinyl chloride polymer or copolymer or polyvinyl butyral, suitably plasticized and pigmented. The plastic compound shall not contain any ingredient that may have an irritating effect on the skin. The compound shall not have a disagreeable odor.

3.2 Construction.

3.2.1 *Type I.*—Type I hospital sheeting shall consist of full width fabric, 3.1.1, coated uniformly on both sides with the rubber compound, 3.1.2. The fabric shall be covered at all points.

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3.2.2 *Type II.*—Type II hospital sheeting shall consist of full width fabric, 3.1.1, coated uniformly on both sides with the plastic compound, 3.1.3. The fabric shall be covered at all points.

3.2.3 *Type III.*—Type III sheeting shall be made from the plastic compound, 3.1.3, and shall not contain fabric.

3.3 Dimensions.

3.3.1 *Thickness.*

3.3.1.1 *Types I and II.*—The thickness of types I and II sheeting shall be not less than 0.015 inch and shall be uniform within 0.002 inch.

3.3.1.2 *Type III.*—The thickness of type III sheeting shall be not less than 0.006- or 0.004-inch as specified in the invitation for bids and shall be uniform within 0.001-inch.

3.3.2 *Width.*

3.3.2.1 *Types I and II.*—A tolerance of plus or minus one inch shall be permitted in the width of types I and II sheeting provided a compensating addition in length is made for any deficiency in width of ½ inch or over.

3.3.2.2 *Type III.*—A tolerance of plus ½ inch will be permitted in the width of type III sheeting. No minus tolerance is permitted.

3.3.3 *Length.*

3.3.3.1 *Types I and II.*—Unless otherwise specified in the invitation for bids, types I and II sheeting shall be furnished in rolls containing 50 to 60 yards. The rolls shall contain not more than 3 pieces and no piece shall be less than 6 yards in length.

3.3.3.2 *Type III.*—Type III sheeting shall be furnished in continuous rolls of the length specified in the invitation for bids.

3.4 Tensile properties.

3.4.1 *Breaking strength, types I and II.*—The breaking strength, grab method, of types I and II sheeting shall be not less than 55 pounds in the warp and not less than 50 pounds in the filling.

3.4.2 *Tensile strength, type III.*—The tensile strength of type III sheeting shall be not less than 2300 pounds per square inch in each direction of the sheeting.

3.4.3 *Ultimate elongation, type III.*—The ultimate elongation of type III sheeting shall be not less than 170 percent in each direction of the sheeting.

3.4.4 *Tearing strength, type III.*—The tearing strength of type III sheeting in any direction shall be not less than 200 pounds per inch of thickness.

3.4.5 *Adhesion of coating, types I and II.*—The adhesion of the coating to the fabric in types I and II sheeting shall be not less than 3.5 pounds per inch of width.

3.5 Resistance to disinfectants.

3.5.1 *Phenol, types I and II sheeting.*—When immersed in a 5 percent solution of phenol as described in 4.3.6.1, types I and II sheeting shall show no evidence of softening, tackiness, hardening, peeling, blistering, or other change that may affect the serviceability of the sheeting; shall not crack at the fold when tested as described in 4.3.6.1.2; and shall show good fastness of color to crocking when tested as described in 4.3.6.1.3.

3.5.2 *Alcohol, types I and II sheeting.*—When immersed in a 70 percent solution of alcohol as described in 4.3.6.2, types I and II sheeting shall show no evidence of softening, tackiness, hardening, peeling, blistering, or other change that may affect the serviceability of the sheeting; shall not crack at the fold when tested as described in 4.3.6.2.1; and shall show good fastness of color to crocking when tested as described in 4.3.6.2.2.

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3.5.3 *Phenol, type III sheeting.* — When immersed in a 5 percent solution of phenol as described in 4.3.6.3, type III sheeting shall show no evidence of softening, tackiness, hardening, or blistering; shall not decrease in weight by more than 4.0 percent; and shall show good fastness of color to crocking when tested as described in 4.3.6.3.1.

3.5.4 *Alcohol, type III sheeting.*—When immersed in a 70 percent solution of alcohol as described in 4.3.6.4, type III sheeting shall show no evidence of softening, tackiness, hardening, or blistering; shall not decrease in weight by more than 4.0 percent; and shall show good fastness of color to crocking when tested as described in 4.3.6.4.1.

3.6 Resistance to accelerated aging, type I sheeting.—When aged as described in 4.3.7.1, type I sheeting shall show no evidence of softening, tackiness, hardening, peeling, blistering, or other change that may affect the serviceability of the sheeting; shall not crack at the fold when tested as described in 4.3.7.1.1; and shall show good fastness of color to crocking when tested as described in 4.3.7.1.2.

3.7 Hydrostatic resistance, types I and II. —Types I and II coated sheeting, shall withstand a pressure of 25 pounds per square inch for a period of 5 minutes when tested as described in 4.3.8.

3.8 Volatile plasticizer.

3.8.1 *Type II.*—The loss in weight of type II sheeting shall not exceed 5 percent and the coating shall show no evidence of softening, tackiness, stiffening, peeling, or blistering when tested as described in 4.3.9.

3.8.2 *Type III.*—The loss in weight of type III sheeting shall not exceed 9.0 and 7.5 percent for class 2 and class 1 sheeting, respectively, and the sheeting shall show no evidence of softening, tackiness, stiffening, or blistering when tested as described in 4.3.9.

3.9 Steam sterilization, types I and II sheeting.—When exposed to steam as described in 4.3.10.1, types I and II sheeting shall show no evidence of softening, tackiness, hardening, peeling, blistering, or other change that may affect the serviceability of the sheeting; shall not crack at the fold when tested as described in 4.3.10.1.1; and shall show good fastness of color to crocking when tested as described in 4.3.10.1.2.

3.10 Workmanship.—The finished sheeting shall have a uniformly smooth, soft finish and the surface shall be free from perforations, uncoated areas, pits, pinholes, blisters, or other defects which may affect the serviceability of the sheeting.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot.*—Unless otherwise specified in the invitation for bids, the lot shall be formed as described in subsection 3 of section 6 of Federal Standard No. 601.

4.1.2 *Inspection.*—Unless otherwise specified in the invitation for bids, the sample for inspection shall be taken as described in subsection 4 of section 6 of Federal Standard No. 601.

4.1.3 *Test.*—Unless otherwise specified in the invitation for bids, the sample for test shall be taken as described in 4.2.3.2.1 of section 6 of Federal Standard No. 601. Sampling plan B of table II of section 6 of Federal Standard No. 601 shall be used.

4.2 Inspection.—Each unit in the sample for inspection shall be inspected for type, class, color, fabric, odor, construction, workmanship, and for dimensions, 4.2.3.1.3.1 in section 6 of Federal Standard No. 601. Six square yards of sheeting shall be unrolled from each sample roll for visual inspection and for determining the width and thickness.

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4.2.1 *Length.*—The length shall be determined as described in procedure 4.2, 4.3, or 4.5 of method 5010 of Federal Specification CCC–T–191. If procedure 4.5 is used, five specimens from each test unit shall be tested.

4.2.2 *Width.*—The minimum and maximum width of the sheeting shall be determined as described in method 2121 of Federal Standard No. 601 except that the scale or tape may be graduated to ⅛ inch.

4.2.3 *Thickness.*

4.2.3.1 *Types I and II sheeting.*—The minimum and maximum thickness of types I and II sheeting shall be determined as described in method 2011 of Federal Standard No. 601.

4.2.3.2 *Type III sheeting.*—The minimum and maximum thickness of the sheeting shall be determined as described in method 2011 of Federal Standard No. 601 except that the gage used shall be graduated to 0.0001 inch, and the value recorded to the nearest 0.0001 inch.

4.3 Tests.

4.3.1 *Breaking strength.* — The breaking strength of types I and II sheeting shall be determined as described in method 5100 of Federal Specification CCC–T–191 except that five specimens from each of the warp and filling directions of each test unit shall be tested. The breaking strength of the test unit shall be the average of the results obtained from the specimens tested in each of the warp and filling directions and shall be recorded separately to the nearest 1.0 pound.

4.3.2 *Tensile strength, type III sheeting.*

4.3.2.1 *Specimen.*

4.3.2.1.1 The specimen shall consist of a strip of sheeting one inch in width and not less than 5 inches in length.

4.3.2.1.2 Care shall be used in cutting the specimen in order to prevent nicks and tears in the edges.

4.3.2.2 *Apparatus.*

4.3.2.2.1 *Testing machine.* — The testing machine described in method 5100 of Federal Specification CCC–T–191 except that the rate of travel of the power actuated grip shall be 20 ± 1 inch per minute and the distance between the jaws of the grips shall be 2 ± ⅛ inch.

4.3.2.2.2 *Dial micrometer.* — The dial micrometer described in method 2011 of Federal Standard No. 601 except that the dial shall be graduated to read to 0.001 inch.

4.3.2.2.3 A scale or tape graduated to 0.01 inch.

4.3.2.2.4 *Cutter.*—A paper cutter or similar apparatus for cutting the specimen to the proper width.

4.3.2.3 *Procedure.*

4.3.2.3.1 The minimum thickness of the specimen shall be determined as described in method 2011 of Federal Standard 601 except that the value shall be recorded to the nearest 0.0001 inch. The minimum width of the specimen shall be determined as described in method 2121 of Federal Standard 601.

4.3.2.3.2 The specimen shall be placed in the grips of the testing machine, care being taken to align the long axis of the specimen with an imaginary line joining the points of attachment of the grips to the machine. The distance between the grips at the start of the test shall be 2 ± ⅛ inch. The grips shall be tightened evenly and uniformly to the degree necessary to hold the specimen without slipping, the machine started, and the force required to rupture the specimen recorded. The tensile strength in pounds per square inch shall be calculated by dividing the force in pounds by the minimum cross-sectional area of the specimen.

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4.3.2.4 *Results.*—Five specimens with the long dimension parallel to the length of the roll and five specimens with the long dimension at right angles to the length of the roll shall be tested. The tensile strength of the test unit shall be the average of the values obtained from the specimens tested in each direction and shall be recorded separately to the nearest 10 pounds per square inch.

4.3.3 *Elongation, type III sheeting.*

4.3.3.1 *Specimen.*—The specimen described in 4.3.2.1.

4.3.3.2 *Apparatus.* — The apparatus described in 4.3.2.2.

4.3.3.2.1 *Dividers.*—A pair of dividers in which the distance between the points can be accurately adjusted to within 0.01 inch.

4.3.3.2.2 *Bench marker.*—A bench marker having two parallel knife edges, ground smooth and true, between 0.002 and 0.003 inch in width at the edge and beveled at angle of not more than 15° to the perpendicular. The distance between the centers of the knife edges shall be 1.000 ± 0.003 inch.

4.3.3.3 *Procedure.*

4.3.3.3.1 Two parallel marks shall be placed symmetrically on the specimen perpendicular to the long axis by means of the bench marker and ink. The marks shall be as fine as practicable consistent with legibility. The specimen shall be placed in the grips of the testing machine. The remainder of the procedure shall be as described in 4.3.2.3.2 except that as the power actuated grip moves, the distance between the two marks shall be followed continuously by means of dividers which shall be used in such a manner as to avoid injuring the specimen. When the specimen ruptures, the dividers shall be placed on the scale and the distance between the pointers read from the scale and the value recorded to the nearest 0.01 inch. The elongation in percent is equal to the distance between the marks on the specimen at the moment of rupture, minus the distance between the knife edges of the bench marker, divided by the distance between the knife edges of the bench marker, and multiplied by 100.

4.3.3.4 *Results.*—Five specimens with the long dimension parallel to the length of the roll and five specimens with the long dimension at right angles to the length of the roll shall be tested. The elongation of the test unit shall be the average of the values obtained from the specimens tested in each direction and shall be recorded separately to the nearest 5 percent.

4.3.4 *Tearing strength, type III sheeting.*—The tearing strength of type III sheeting shall be determined as described in method 4211 of Federal Standard No. 601. A specimen cut with die C shall be used.

4.3.5 *Adhesion of coating, types I and II.*—The adhesion of the coating to the fabric of types I and II sheeting shall be determined as described in method 8211 of Federal Standard 601.

4.3.6 *Resistance to disinfectants.*

4.3.6.1 *Phenol, types I and II.*—The resistance of types I and II sheeting to a solution of phenol shall be determined as described in method 6711 of Federal Standard No. 601 with the following exceptions: (1) The specimen shall be 8 by 4 inches. (2) Two specimens with the long dimension parallel to the length of the sheeting and two specimens with the long dimension at right angles to the long dimension of the sheeting from each test unit shall be tested. (3) The specimen shall be examined for evidence of softening, tackiness, stiffening, peeling, or blistering of the coating.

4.3.6.1.2 Twenty-four hours after the end of the immersion period, one specimen from

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each direction of each test unit shall be tested for cracking as follows: The specimen shall be folded double and placed on a hard, smooth, flat surface. The folded edge shall be rolled 10 times, 5 in each direction, along its full length with a metal roller approximately one inch in diameter, under a load of 5 pounds. Each specimen shall be examined immediately for cracking at the fold.

4.3.6.1.3 Twenty-four hours after the end of the immersion period, each specimen in each direction of each test unit shall be tested for resistance to crocking as described in method 5650 of Federal Specification CCC–T–191, and the resistance rated as good, fair, or poor.

4.3.6.2 *Alcohol, types I and II.*—The resistance of types I and II sheeting to a solution of alcohol shall be determined as described in 4.3.6.1 except that (1) the immersion media shall be a 70 percent solution of alcohol and (2) the time of immersion shall be 22 ± ¼ hour.

4.3.6.2.1 Twenty-four hours after the end of the immersion period, one specimen from each direction of each test unit shall be tested for cracking as described in 4.3.6.1.2.

4.3.6.2.2 Twenty-four hours after the end of the immersion period, each specimen in each direction of each test unit shall be tested for resistance to crocking as described in method 5650 of Federal Specification CCC–T–191, and the resistance rated as good, fair, or poor.

4.3.6.3 *Type III sheeting—phenol.*—Resistance of type III sheeting to a solution of phenol shall be determined as described in method 6711 of Federal Standard No. 601, with the following exceptions: (1) The specimen shall be 8 by 4 inches. (2) One specimen with the long dimension parallel to the length of the sheeting and one specimen with the long dimension at right angles to the length of the sheeting from each test unit shall be tested. (3) The specimen shall be weighed before immersion. (4) At the end of the immersion period, the specimen shall be removed from the immersion medium, gently wiped with a soft absorbent cloth or tissue, examined for evidence of softening, tackiness, hardening, or blistering, and immediately weighed, and the percentage change in weight based on the original weight calculated.

4.3.6.3.1 Twenty-four hours after the end of the immersion period, each specimen shall be tested for resistance to crocking as described in method 5650 of Federal Specification CCC–T–191, and the resistance rated as good, fair, or poor.

4.3.6.4 *Resistance to alcohol, type III.*—The resistance of type III sheeting to a solution of alcohol shall be determined as described in 4.3.6.3 except that (1) the immersion media shall be 70 percent solution of alcohol and (2) the time of immersion shall be 22 ± ¼ hour.

4.3.6.4.1 Twenty-four hours after the end of the immersion period, each specimen shall be tested for resistance to crocking as described in method 5650 of Federal Specification CCC–T–191, and the resistance rated as good, fair, or poor.

4.3.7 *Resistance to accelerated aging.*

4.3.7.1 Type I sheeting shall be exposed to circulating air as described in method 7221 or 7231 of Federal Standard 601 with the following exceptions: (1) The specimen shall be 8 by 4 inches. (2) Two specimens with the long dimension parallel to the length of the sheeting and two specimens with the long dimension at right angles to the long dimension of the sheeting from each test unit shall be tested. (3) The specimens shall be examined for evidence of softening, tackiness, hardening, peeling, or blistering of the coating.

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4.3.7.1.1 Twenty-four hours after the removal from the aging chamber, one specimen from each direction of the sheeting shall be tested for cracking as described in 4.3.6.1.2.

4.3.7.1.2 Twenty-four hours after removal from the aging chamber, one specimen from each direction of the sheeting shall be tested for resistance to crocking as described in method 5650 of Federal Specification CCC–T–191, and the resistance rated as good, fair, or poor.

4.3.8 *Hydrostatic resistance, types I and II.* —The resistance of types I and II sheeting to hydrostatic pressure shall be determined as described in method 10511 of Federal Standard No. 601, procedure B. The pressure shall be 25 pounds per square inch.

4.3.9 *Volatile plasticizer.* — The volatile plasticizer in types II and III sheeting shall be determined as follows:

4.3.9.1 *Specimen.*—The specimen shall consist of disks 2 inches in diameter. Three specimens from each test unit shall be tested.

4.3.9.2 *Apparatus.*

4.3.9.2.1 *Balance.*—An accurate analytical balance equipped with class S weights or better.

4.3.9.2.2 *Oven or bath.*—A thermostatically controlled oven or bath capable of maintaining the required temperature to within ± 1.0°C. (1.8°F.).

4.3.9.2.3 *Containers.*—One-pint metal paint cans, 3⅜ inch in diameter, or 1-pint Mason jars of approximately the same diameter.

4.3.9.2.4 *Activated carbon.*—Columbia Activated Carbon, grade AC, 6/14 mesh, manufactured by Carbide and Carbon Chemicals Corporation, or equivalent, shall be used. Care should be taken that an airtight storage container is used for the activated carbon, and that fresh material is used for each test, unless it can be shown that re-use does not affect the results.

4.3.9.3 *Procedure.*

4.3.9.3.1 The specimen shall be preconditioned for 40 hours at a temperature of 23° ± 1.1°C. (73.5° ± 2°F.) and a relative humidity of 50 ± 4 percent. Adequate air circulation shall be provided on all sides of the specimen by placing them in suitable racks, hanging them from metal clips, or laying them on wide-mesh wire screen frames with at least one inch between the screen and the surface of the bench.

4.3.9.3.2 After the specimen has been conditioned in accordance with 4.3.9.3.1, it shall be weighed on the analytical balance, and the weight designated as W_1.

4.3.9.3.3 Only specimens of the same composition of formulation shall be tested in a single container, because of the possibility of cross migration between compositions having varying plasticizer types or concentrations, or both.

4.3.9.3.4 One hundred and twenty cubic centimeters of activated carbon shall be spread evenly on the bottom of the container. The specimen shall be placed on top of the activated carbon and covered with 120 cm^3 of the carbon. A second specimen shall be placed on top of the first and covered with 120 cm^3 of the carbon, followed by a third specimen, and then 120 cm^3 more of activated carbon. A cover shall be placed on the container in such a manner that the container will be vented. This is necessary in order that possible pressure build-up in the container during heating will be relieved. Care shall be taken that in no case the carbon will be packed by pressure other than in the weight of the composite sandwich in the container.

4.3.9.3.5 The container shall be placed upright in the oven or bath.

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4.3.9.3.6 The temperature of the oven or bath shall be 70.0° ± 1.0° C. (158.0° ± 1.8° F.) and the duration of the test shall be 22 ± ¼ hours.

4.3.9.3.7 At the end of the 22 hour period, the container shall be removed from the oven and allowed to cool to room temperature. The specimens shall then be removed from the container, brushed free of carbon, and reconditioned in accordance with 4.3.9.3.1.

4.3.9.3.8 After the specimen has been reconditioned, it shall be reweighed on the analytical balance and the weight recorded as W_2.

4.3.9.3.9 After the specimen has been weighed, it shall be examined for evidence of softening, tackiness, stiffening, peeling, or blistering.

4.3.9.4 *Results.*

4.3.9.4.1 *Calculation.*—The plasticizer volatility, expressed as percentage weight loss as compared to the original specimen weight, shall be calculated as follows:

$$\text{Weight loss, percent} = \frac{W_1 - W_2}{W_1} \times 100$$

where:

W_1 = initial weight of test specimen.

W_2 = final weight of test specimen.

4.3.9.4.2 The average value obtained for the three specimens shall be recorded to the nearest 0.1 percent as volatile plasticizer of the test unit.

4.3.10 *Steam sterilization, types I and II.*

4.3.10.1 Types I and II sheeting shall be subjected to steam sterilization as described in method 7411 of Federal Standard No. 601 except that: (1) The specimen shall be a piece of sheeting 12 inches by 12 inches folded once to give a rectangle 12 by 6 inches and folded again to give a square 6 by 6 inches. (2) The specimen shall be examined for softening, hardening, tackiness, peeling, and blistering. (3) One specimen from each test unit shall be tested.

4.3.10.1.1 Twenty-four hours after the end of the fifth period of sterilization, one piece 8 by 6 inches shall be cut from each direction of the sterilized specimen and tested for resistance to cracking as described in 4.3.6.1.2.

4.3.10.1.2 Twenty-four hours after the end of the fifth period of sterilization, one piece of sheeting 8 by 6 inches shall be cut from each direction of the sterilized specimen and tested for resistance to crocking as described in method 5650 of Federal Specification CCC–T–191 and the resistance rated as good, fair, or poor.

4.4 **Compliance criteria.** — Compliance of the lot shall be based on the principles described in section 6 of Federal Standard No. 601.

5. PREPARATION FOR DELIVERY

5.1 Nonmilitary.

5.1.1 *Packaging.*—Unless otherwise specified in the invitation for bids, types I and II sheeting shall be packaged in rolls containing 50 to 60 yards. Type III sheeting shall be packaged in rolls of the length specified in the invitation for bids.

5.1.1.1 Unless otherwise specified in the invitation for bids, each roll shall be wrapped in paper and sealed.

5.1.2 *Packing.*—Unless otherwise specified in the invitation for bids, the subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation at the lowest rate, to the point of delivery.

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5.1.3 *Marking.*

5.1.3.1 *Packages.*—Unless otherwise specified in the invitation for bids, each unit package shall be marked with the name of the commodity, type, class, and the quantity contained therein, and name of contractor.

5.1.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the commodity, type, class, quantity contained therein as defined by the contract or order under which shipment is made, the name of the manufacturer or contractor, the number of the contract or purchase order, and the gross weight.

5.2 Military agencies.

5.2.1 *Army and Navy.*

5.2.1.1 *Packaging and packing.*

5.2.1.1.1 *Level A and Level B.*—Sheeting shall be packaged and packed in accordance with the requirements of Military Specification MIL–P–12439.

5.2.1.1.2 *Level C.*—Sheeting shall be packed in containers, which (together with packaging, wrapping, and inner packing) are acceptable by common or other carriers for safe transportation to point of destination specified in shipping instructions, at the lowest transportation rate for such items.

5.2.2 *Air Force.*

5.2.2.1 *Packaging.*

5.2.2.1.1 *Level A.*—Unless otherwise specified in the invitation for bids, types I and II sheeting shall be packaged in rolls containing 50 to 60 yards. Type III sheeting shall be packaged in rolls of the length specified in the invitation for bids. Each roll shall be wound on a substantial core with an inside diameter of not less than 3 inches and shall be suitably restrained from unwinding. Rolls of sheeting shall be individually wrapped with two thicknesses of kraft paper suitably secured, and overwrapped with waterproof barrier-material conforming to Military Specification Jan–P–125, type C–1 or C–2, with all folds, seams, and laps sealed with water-resistant adhesive conforming to Military Specification MIL–A–140.

5.2.2.1.2 *Level B.*—The rolls shall be packaged as specified in 5.2.2.1.1 except that the waterproof overwrap shall be omitted.

5.2.2.1.3 *Level C.*—The rolls shall be packaged in conformance with the manufacturer's commercial practice.

5.2.2.2 *Packing.*

5.2.2.2.1 *Level A.*—Unit packaged rolls as specified in 5.2.2.1.1 and of like description shall be packaged in quantities not to exceed 200 pounds gross weight in snug-fitting shipping containers constructed, closed, and strapped in conformance with Federal Specifications PPP–B–601, PPP–B–621, PPP–B–585, or Military Specification MIL–B–138, unless a single roll of sheeting exceeds this weight limitation.

5.2.2.2.2 *Level B.*—Unit packaged rolls as specified in 5.2.2.1.2 and of like description shall be packed in snug-fitting shipping containers constructed and closed in conformance with Federal Specification PPP–B–601, PPP–B–636, PPP–B–591, or PPP–B–585.

5.2.2.2.3 *Level C.*—The rolls packaged as specified in 5.2.2.1.3 shall be packed in containers which are acceptable by common or other carriers for safe transportation to point of destination specified in shipping instructions, at the lowest transportation rate for such supplies.

5.2.2.3 *Marking for identification.*—Unless otherwise specified in the invitation for bids, each unit packaged roll shall be marked with the name of the commodity, type, class, and the quantity contained therein, and name of contractor.

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5.2.3 *Marking for shipment.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Grade, type, width, class, and color (1.2).

(*c*) Length (3.3.3).

6.1.1 *Military only.* — Procurement documents should specify the applicable levels of packaging and packing required. (See 5.2.)

6.2 It is not intended that type III sheeting be exposed to steam sterilization.

6.3 Transportation description.—Transportation description applicable to this item is:

Type I

Cloth, rubber coated, cotton, not otherwise indexed by name, other than artificial or imitation leather.
Carload minimum weight 30,000 pounds.

Cloth, cotton, coated with rubber.
Truckload minimum weight 30,000 pounds.

Types II and III

Dry goods, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Cloth, cotton, coated with plastic.
Truckload minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—M Q
Navy—Md
Air Force.

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